Nowcasting Italian GDP

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Abstract

This study investigates short-term forecasting of the third quarter of 2019 Italian GDP through a real-time nowcasting framework, pretending to be in different points in time (September, October, November 2019). Using official ISTAT data vintages available at the time of estimation, the analysis reproduces the informational environment faced by policymakers, ensuring that results are unaffected by ex-post data revisions. The main objective is to assess the predictive performance of alternative econometric models that integrate both quantitative and qualitative indicators to capture the short-term dynamics of economic activity.

The empirical strategy combines hard and soft indicators to reflect different aspects of the business cycle. Industrial production acts as the principal quantitative predictor, while qualitative survey indicators are summarized through Principal Component Analysis (PCA) to extract latent factors. Several model classes are compared, including Bridge Models and multiple Mixed Data Sampling (MIDAS) specifications, such as Exponential Almon, polynomial distributed lags (PDL1, PDL2), unrestricted (U-MIDAS), and GETS-MIDAS.

In the end, we obtain one forecast per model. When comparing them with our official target, we notice a large discrepancy across vintages in scale. To make the results more comparable, we instead analyze the quarter-over-quarter variation. Since a single forecast is too limited to draw robust conclusions, we suggest using additional model evaluation methods, such as the calculation of the Mean Squared Error (MSE).

Data

The empirical analysis relies on a real-time dataset that replicates the information available to policymakers in the third quarter of 2019. Using real-time vintages, as released by Istat, ensures that the analysis reflects the data environment faced by decision-makers at that time and avoids the bias arising from subsequent data revisions. The target variable is Italy's GDP, measured at quarterly frequency and expressed in constant prices to capture the evolution of real economic activity and to reflect the total value of goods and services produced within the national economy. As a higher-frequency indicator, the Industrial Production Index (IPI) is included. The IPI, published monthly by Istat, tracks short-term variations in industrial output, including manufacturing, mining, and utilities, and serves as a timely measure of real economic activity. Since both GDP and IPI are found to be integrated of order one, I(1), they are transformed by first differencing; in particular, GDP is expressed in log-differenced form to approximate quarterly growth rates, ensuring stationarity and meaningful inference.

In addition to this quantitative hard indicator, the dataset incorporates qualitative "soft" indicators obtained from ISTAT's monthly business and consumer (Table 3). Since these indicators are numerous and often correlated, they are summarized through a Principal Components Analysis (PCA).

By combining quarterly GDP data, monthly industrial production and the principal components extracted from qualitative surveys, the dataset provides a comprehensive and balanced representation of Italy's short-term economic dynamics.

Models

AutoARMA

A simple benchmark for forecasting can be provided by the ARMA model (Figure 1). We use the `AutoARMA` function, which automatically selects an ARMA(1,1) specification. The MA component has a unit root, which is typical in the presence of over-differencing: in fact, GDP was differenced twice. However, since the MA term is

not statistically significant, we focus on the autoregressive part, which has a 0.46 coefficient. The model yields an $R^2 = 0.26$, relatively low compared to other specifications.

Bridge Model

The first model we employ is a simple *Bridge Model*, which allows us to "bridge" high- and low-frequency data. The general formulation is given by:

$$\hat{y}^L_{T+1|T+1-(m-i)/m} = \hat{a}_T + \hat{b}_T \sum_{j=1}^{i} a_{m-j}\, x^H_{T+1-(m-j)/m} + \hat{b}_T \sum_{h=1}^{i-m} a_{\,m-(i-h)}\, \phi_h\big(L^{1/m}\big)\, x^H_{T+1-(m-1)/m},$$
$$i = 1, \ldots, m-1. \quad (1)$$

The model works by aggregating high-frequency data (inducing a loss of variability). In our case, we use the Industrial Production Index (IPI) as a monthly hard indicator. It is aggregated to the quarterly frequency by taking the mean. If, as in our case when we nowcast in September and October, we do not have all three monthly observations that make up the quarter, we forecast the missing entries with a simple AR model. Using the `AutoARMA` function, an ARMA(1,2) was selected, with an autoregressive coefficient of 0.74. This value is almost the same for each of the three cases we have. In our empirical application, we estimate a Bridge Model where GDP is the dependent variable, and the explanatory variables are GDP_{t-1}, IPI_t, and IPI_{t-1}.

We find that the constant term is statistically significant, corresponding to an average baseline GDP growth of approximately 0.2%. The past value of GDP is also significant, although only moderately, indicating that past growth tends to persist.

Interestingly, only the current value of the IPI is statistically significant, implying that past quarter's IPI values have little explanatory power. The Durbin–Watson statistic suggests that there is no autocorrelation in the residuals (Figure 5).

For all vintages, we obtain almost the same coefficient estimates. This confirms the stability of the model across data vintages.

The forecasted GDP value for each vintage is:

Table 1: Summary of GDP Forecasts by Data Vintage

Model	IPI Version	GDP Forecast
GDP_BM_1	Vintage 1	404,289.3
GDP_BM_2	Vintage 2	404,227.0
GDP_BM_3	vintage 3	404,222.7

As we can observe, the forecasts produced by the successive Bridge Models decrease slightly as new information becomes available. This suggests that industrial production deteriorated in the last two months of the quarter, leading to a downward update of GDP growth expectations comapred to our forecasted missing entries.

Table 2 reports the monthly values of the Industrial Production Index (IPI) during Q3, where the first two columns contain also forecasted missing values.

Table 2: Monthly and Quarterly Industrial Production Index (IPI) — 2019Q3

Period	**Vintage 1**	**Vintage 2**	**Vintage 3**
2019M07	104.6900	104.5900	104.5900
2019M08	104.9559	104.8900	104.9900
2019M09	104.8800	104.8015	104.5900
2019Q3	**104.84**	**104.76**	**104.72**

As we can observe, as more real data become available, the value of the IPI gradually decreases. This downward update directly affects our GDP nowcasts through the Bridge Model mechanism (that uses the mean of the three months).

MIDAS

The next step is to consider MIDAS as a methodological evolution of Bridge models, given their greater flexibility and dynamism. The objective of both models is the same, namely to predict a low-frequency variable using high-frequency indicators. In the case analysed, the objective is GDP, which has a quarterly frequency, using a series of monthly indicators (soft or hard). The main difference between these two models lies in the technique used to incorporate high-frequency information into the model: MIDAS models use high-frequency data directly without any aggregation, but include a parametric weighting function that determines how each lag of the high-frequency variable contributes to the explanation of the target variable:

$$y_t = \alpha + \rho y_{t-1} + \beta \sum_{k=0}^{K} w(k;\theta) x_{t,k} + \varepsilon_t$$

where $w(k;\theta)$ is generally a Beta or Almon type function, parameterised by a few coefficients θ that define the trend of the weights over time, and ρy_{t-1} represents the autoregressive component that captures the dynamic dependence of the target variable In this way, the model is able to efficiently use all the information contained in the high-frequency data, allowing the data itself to determine which past observations are most relevant. A further extension is provided by U-MIDAS (Unrestricted MIDAS), which dispenses with the parametric specification of weights, freely estimating a distinct coefficient for each lag:

$$y_t = \alpha + \rho y_{t-1} + \sum_{k=0}^{K} \gamma_k \, x_{t,k} + \varepsilon_t$$

This formulation allows greater flexibility in representing the relationship between the two frequencies, but increases the number of parameters to be estimated. To avoid overfitting problems, especially when high-frequency variable lags are also introduced, the parametric weight functions mentioned above are often used in practice. Furthermore, compared to Bridge models, MIDAS models allow for the introduction of an autoregressive term, as well as enabling the estimation of endogenous weights on high-frequency regressors, improving forecasting capacity and correctly capturing all the information from high-frequency regressors.

Exponential ALMON

The results, obtained using an Exponential Almon weighting function, show an increase in the autoregressive coefficient: from 0.27 in the September model to approximately 0.37 in the November one. This indicates greater persistence in GDP dynamics once more recent information is incorporated. From the perspective of the weight specification, the parameters of the Exponential Almon function are not significant in the first two estimations, suggesting an essentially flat weight distribution: in September and October, the three monthly IPI observations contribute similarly to GDP forecasting. In the November model, however, both parameters become statistically significant ($p < 0.05$), and the weight function exhibits a well-defined curvature. Specifically, the weights are $w_0 = 0.1138$, $w_1 = 0.1638$ and $w_2 = 0.1064$, forming a pattern with a peak in the intermediate month. This suggests that the impact of industrial production on GDP is not immediate but reaches its maximum after about one month, while information that is either too recent or too distant exerts a weaker influence. The explanatory power of the model progressively improves across the three nowcasts: the Adjusted R^2 increases and the AIC decreases, indicating a higher goodness of fit. Moreover, the Durbin–Watson statistic remains close to 2, signaling the absence of residual autocorrelation and a well-specified model (Table 6, Table 7, Table 8). Overall, the results confirm that updating high-frequency data allows the MIDAS model to gradually refine the weight distribution. Empirical evidence therefore suggests that, as nowcasting progresses through the months, the model becomes increasingly informed and stable, capturing more accurately the contribution of industrial production to quarterly GDP growth.

PDL1 ALMON

Using a first-degree Almon polynomial function, the weights associated with the three monthly lags of the IPI follow a linear relationship of the form $w_k = a_0 + a_1 k$. However, with three lags, the estimation produced a zero slope, indicating no variation in the weights over time: consequently, the three IPI months influence GDP with constant weights (very similar across the three nowcasts, with values around 0.122). This flat distribution suggests that, under the linear specification, the model interprets the effect of the IPI as uniform, with no prevailing temporal delays. It is therefore observed that, as the nowcast progresses, the data updates improve the overall precision of

the estimation but do not alter the timing with which industrial production affects GDP. As in the previous case, the autoregressive component shows a gradual increase, rising from 0.285 to 0.365. This again highlights that GDP persistence is a key factor in explaining its dynamics. The same applies to the model's goodness-of-fit indicators, which improve over the analyzed period, consistent with the idea that the inclusion of monthly data enhances forecasting performance (Table 9, Table 10, Table 11). In summary, when comparing September, October, and November, the PDL1 model exhibits remarkable structural stability. However, the shape of the weights indicates that the first-degree linear specification fails to capture significant temporal differences in the impact of the various IPI months, unlike the Exponential Almon model.

PDL2 ALMON

The use of a second-degree Almon polynomial (PDL2) allows for a more flexible representation of the weight distribution compared to the linear PDL1 model, enabling the MIDAS framework to capture potential curvature in the temporal relationship between the IPI and GDP growth. Overall, the results display a consistent and gradual evolution over time, with improvements in explanatory power and a slight change in the shape of the weights. As in the previous specifications, the autoregressive component of the model increases steadily from 0.281 to 0.364 between September and November, indicating a stronger dependence of GDP growth on past values as more recent information is incorporated. In September, the weight function shows an upward trend, suggesting that more distant months (lag 2) are more informative for GDP growth; the impact of industrial production therefore appears to be delayed. In October, the shape of the weights reverses, becoming downward-sloping: the most recent IPI observation (lag 0) gains importance, indicating a more immediate effect of industrial production on GDP. Finally, in November, the distribution tends to flatten: the three months display similar weights, with a slight predominance of more recent values, forming an almost flat pattern consistent with a balance between contemporaneous and lagged effects. This evolution in the shape of the weights suggests that the PDL2 model is capable of adapting over time: when information is still limited (September), it tends to assign greater relevance to older lags to compensate for the lack of updated data; as new data become available (October and November), the emphasis shifts toward more recent months. Although the quadratic component of the polynomial (PDL2) is not statistically significant in any of the three nowcasts, the change in the slope of the weights over time highlights that the model responds coherently to data updates, shifting from a more delayed effect in the early stages to a more immediate one in the later periods (Table 12, Table 13, Table 14). In summary, when comparing the three nowcasting exercises, the PDL2 model exhibits greater flexibility than PDL1, capturing subtle changes in the temporal structure of the IPI's effect on GDP.

U-MIDAS

The U-MIDAS represents the nonparametric version of the MIDAS model, in which the coefficients associated with the different lags of the high-frequency variable are freely estimated, without imposing any functional form on the weights. This specification allows the model to fully adapt to the data, capturing possible irregularities in the temporal distribution of the IPI's effect on GDP growth. The autoregressive component increases in line with what is observed in the other specifications. The most interesting aspect concerns the distribution of the weights across the three monthly lags of the IPI, which, in the U-MIDAS framework, are estimated individually. In September, the weight distribution displays a slight bell-shaped pattern with a peak at the second lag (the intermediate month), indicating that the IPI effect materializes with about a one-month delay. In October, the shape reverses and becomes downward-sloping: the most recent IPI month becomes the most influential, suggesting a more immediate impact. Finally, in November, the distribution once again takes on a bell-shaped form, this time more pronounced, confirming that the maximum effect of the IPI occurs in the intermediate month, while both very recent and more distant months are less influential (Table 15, Table 16, Table 17). This evolution in the weights deserves attention from an economic perspective, as it is a pattern also observed in the other models. At the beginning of the quarter (September), when only partial data are available, the model tends to assign greater relevance to earlier information. As new data are released (October), more recent information gains importance. Finally, when nearly complete data become available (November), the model converges toward a more balanced and curved distribution, which realistically reflects the temporal transmission of industrial activity to GDP. The progressive improvement in model quality suggests that U-MIDAS benefits more from data updates than the parametric-weight models, as it can freely adapt to monthly information changes without structural constraints. In summary, the sequential comparison across the three nowcasting rounds shows that: the relationship between IPI and GDP remains positive and significant throughout the period, the weight pattern evolves from a mild bell shape (September) to a decreasing form (October) and finally to a well-defined bell shape (November); the model's explanatory power also improves consistently, reaching its best performance in the final November

nowcast. Overall, the U-MIDAS model stands out for its greater flexibility and its ability to more faithfully reflect the temporal sequence through which industrial activity influences GDP. It is worth noting that, in this case, the U-MIDAS produces the same results as the PDL3 model. This occurs because a third-degree polynomial can always be fitted through any three points. Consequently, when PDL3 estimates the weights, it no longer imposes any particular functional form but selects exactly the same values estimated by the unrestricted U-MIDAS. For this reason, the analysis includes the U-MIDAS model rather than PDL3.

GETS MIDAS

The GETS (General-to-Specific) methodology is an automated variable selection procedure that starts from a general unrestricted model and sequentially eliminates statistically insignificant variables while ensuring the validity of diagnostic tests at each step. This approach allows for a systematic reduction of model complexity while preserving statistical adequacy and empirical coherence. In this framework, the final specification retains only the variables that provide significant explanatory power and satisfy the underlying model assumptions. Here we estimated three MIDAS models using the GETS procedure, one for each month (Table 18). In this case, GETS is not used to select the variables, but rather to select the optimal lags of the IPI indicator. All the estimated models display positive and significant coefficients for both the lagged GDP growth term and the lags of IPI, as expected. The interesting result concerns the selection of lags by the GETS method. We note that in the first model, where less information is available, GETS retains more distant lags (LAG 7, LAG 16). Then, as more information becomes available, only the more recent lags of IPI remain. Moreover, another interesting result regards the model fit. We can see that as the available information increases, the fitting improves as well. The R^2 rises (from approximately 0.45 to 0.50), the AIC decreases, and the log-likelihood increases. The Durbin–Watson statistic indicates no autocorrelation in the residuals, which is expected since the models are estimated through the GETS procedure, ensuring that diagnostic tests are satisfied.

PCA

The Principal Component Analysis was conducted on 32 standardized soft indicators using the correlation matrix. The first principal component alone explains 36.9% of the total variance, while the first five components account for approximately 69% of the overall variability in the data. This suggests the existence of a strong common component underlying most indicators.

UNPC models

Now, we use the first 5 principal components, together with the IPI, to estimate a U-MIDAS, one for each month (Figure 2). In all three U-MIDAS models, we observe a high R^2, always above 0.8. The IPI still exerts the strongest effect on GDP, as its coefficients are much larger than those associated with the principal components. The magnitude of this effect seems to decrease as we move closer to the nowcast (i.e., as more data become available). The principal components generally display small coefficients. Among them, the first component (PC1) has the largest impact, which is consistent with the fact that it explains the highest share of variability. Almost all coefficients, however, appear individually insignificant despite the high overall explanatory power. This likely results from the limited sample size and the large number of parameters to be estimated. Since soft indicators start only around the year 2000, the GDP sample is shortened (ragged edges problem), leaving relatively few observations. Consequently, large standard errors combined with small coefficient magnitudes lead to statistically insignificant estimates. Nevertheless, the model fits the dynamics of GDP growth quite well. Given the forecasting purpose of the analysis, it is more relevant to focus on predictive performance rather than on statistical inference.

GEPC models

Now, we apply the GETS selection to the previous models, estimating MIDAS specifications that include both the IPI and the principal components, with the GETS procedure determining which lags (and therefore which variables) to retain (Table 19). The first important result is that the IPI is always included in the model, as it represents a hard indicator with strong explanatory power. However, we can observe that not all the lags are always selected (LAG1, which is excluded in the second model). Regarding the principal components, we find that PC1 is always retained (though with only one lag), since it explains the largest share of GDP variability. Its coefficient is positive, thus contributing to the explanation of GDP growth; however, being a principal component, it is not directly interpretable in economic terms. The other PCs are, in some cases, selected and in others excluded. In the first model PC3, PC3, PC5 are included; in the second model, PC3 and PC5 are retained; while in the third model PC2 and PC5 are included, each with different lag selections. Concerning model fit, both specifications

display a high adjusted R^2 (higher than 0.76). It can be observed that, in this case, the fit does not significantly improve as we move closer to the nowcast; in fact, the AIC remains very similar across models, as does the log-likelihood.

Results and Discussion

Table 20 reports our forecast results. In all cases, the nowcast in November is lower than the nowcast made in September. This outcome reflects the new releases of October and November, which show a decreasing trend in the IPI series. For most models (except the Bridge Model, GETS, and UMIDAS), the October forecast is lower than both the September and November estimates. This pattern arises from differences in model mechanisms. Interestingly, result from PDL1 and Stepwise are identical. This occurs because, with only three monthly lags of the IPI, the linear restriction imposed by the PDL1 model produces coefficients that are nearly constant over time: the three months of industrial production end up having a uniform weight on GDP dynamics. Similarly, the stepwise procedure, which automatically selects the most informative lags, finds no significant differences among the months and therefore tends to retain only a single average effect of the IPI. As a result, both specifications summarize the impact of the IPI on GDP through one stable coefficient, leading to practically identical estimates and forecasts.

We are left with the issue of comparing those results. At first glance, when comparing the levels of our forecasts, we notice a substantial discrepancy, approximately a 6% difference, between our predicted GDP level and the official target. However, there is a plausible explanation for this phenomenon.

Every five years (usually in that ending by 4 and 9), the Italian National Institute of Statistics (ISTAT), in coordination with other European statistical agencies, conducts a comprehensive "General Revision of National Accounts." In the 2019 and 2024 vintages, in fact, we observe a jump in the scale of GDP due to changes in the reference base year and methodological updates. [1]

These revisions aim to:

- Update the base year of chain-linked volumes (e.g., from 2010 to 2015, and then to 2021).
- Integrate new statistical sources such as business balance sheets, public accounts, and structural surveys.
- Enhance consistency with European standards (ESA 2010) and prepare for future methodological improvements.

Such benchmark revisions raise the issue of data vintages: different releases of the same variable may not be directly comparable, as their underlying scale and definitions evolve over time.

Because of these revisions, directly comparing forecast levels may lead to misleading conclusions. Instead, we compute the *percentage change* relative to the previous period (Table 21). This approach allows us to assess whether the dynamics (growth rates) are consistent with our target values, even if the absolute levels differ. Results are very different from model to model and we cannot say which is the best.

One potential improvement could be to *pool* forecasts obtained from different vintages or model specifications. Pooling can often enhance forecast accuracy. By taking the mean of forecast made in November (+ AutoARMA) we get a value of 404118.3 milion euros. It is a 0.08% growth with respect to Q2 and close to the 0.06% of the official target.

With only one forecast per model, drawing firm conclusions is difficult. An hypothesis of work is that of carrying out the study of MSE by dividing the sample into training and testing to evaluate models. Moreover, we should verify whether comparing our forecasts to the revised GDP series is conceptually meaningful, since it incorporates new data sources and definitions.

One thing we can be sure of is that the forecasts made in November have to be the most reliable (having more information). In fact, we can see that GDP target grew by 0.06% and there are some models that predict an increase of 0.25% in semptember. As new data become available, however, the forecasts tend to adjust downward, incorporating signals from the incoming information that the economic cycle is deteriorating.

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[1] For further details, see: `https://www.istat.it/comunicato-stampa/la-revisione-generale-dei-conti-nazionali-2024-anni-1995-2023`.

List of Tables

Table 3: Soft indicators used in the analysis

Category	Indicators
Manufacturing	Manufacturing business confidence climate (index; base 2021=100).
	Assessments on orders, domestic and foreign orders, production level, and finished product inventories.
	Liquidity level vs. operational needs.
	Expectations on orders, production, selling prices, general economy, liquidity, and employment.
Construction	Construction business confidence climate (index; base 2021=100).
	Assessments on order books and construction plans.
	Assessments on construction activity (past 3 months).
	Trends in orders, prices, and employment (next 3 months).
Services	Services business confidence climate (index; base 2021=100).
	General business trend.
	Orders and demand (current and expected).
	Expected employment and selling prices.
	Employment assessments.
	Expected Italian economic trend.
Consumers	Consumer confidence climate (index; base 2021=100).
	Personal, economic, current, and future consumer climate (weighted estimates).

Dependent Variable: DLOG(GDP,2)
Method: ARMA Maximum Likelihood (BFGS)
Date: 10/18/25 Time: 16:20
Sample: 1980Q1 2019Q2
Included observations: 158
Convergence achieved after 29 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-5.18E-05	2.33E-05	-2.217940	0.0280
AR(1)	0.459637	0.057557	7.985821	0.0000
MA(1)	-0.999993	3794.019	-0.000264	0.9998
SIGMASQ	3.40E-05	0.006926	0.004909	0.9961

R-squared	0.263510	Mean dependent var	-0.000145
Adjusted R-squared	0.249163	S.D. dependent var	0.006816
S.E. of regression	0.005906	Akaike info criterion	-7.374747
Sum squared resid	0.005372	Schwarz criterion	-7.297213
Log likelihood	586.6050	Hannan-Quinn criter.	-7.343259
F-statistic	18.36669	Durbin-Watson stat	1.987897
Prob(F-statistic)	0.000000		

Inverted AR Roots	.46
Inverted MA Roots	1.00

Figure 1: GDP AutoARMA

Dependent Variable: DLOG(IPI_237)
Method: ARMA Maximum Likelihood (BFGS)
Date: 10/18/25 Time: 16:07
Sample: 1980M02 2019M07
Included observations: 474
Convergence achieved after 23 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000302	0.000866	0.348799	0.7274
AR(1)	0.746852	0.077118	9.684580	0.0000
MA(1)	-1.095340	0.078219	-14.00351	0.0000
MA(2)	0.388736	0.036339	10.69747	0.0000
SIGMASQ	0.000244	1.02E-05	23.99694	0.0000

R-squared	0.123659	Mean dependent var	0.000319
Adjusted R-squared	0.116185	S.D. dependent var	0.016693
S.E. of regression	0.015693	Akaike info criterion	-5.460105
Sum squared resid	0.115503	Schwarz criterion	-5.416211
Log likelihood	1299.045	Hannan-Quinn criter.	-5.442842
F-statistic	16.54490	Durbin-Watson stat	1.961060
Prob(F-statistic)	0.000000		

Inverted AR Roots	.75	
Inverted MA Roots	.55-.30i	.55+.30i

(AutoARMA vintage 1)

Dependent Variable: DLOG(IPI_238)
Method: ARMA Maximum Likelihood (BFGS)
Date: 10/18/25 Time: 16:09
Sample: 1980M02 2019M08
Included observations: 475
Convergence achieved after 18 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000300	0.000864	0.346939	0.7288
AR(1)	0.746857	0.077145	9.681213	0.0000
MA(1)	-1.095243	0.078257	-13.99545	0.0000
MA(2)	0.388432	0.036308	10.69829	0.0000
SIGMASQ	0.000243	1.01E-05	24.02529	0.0000

R-squared	0.123383	Mean dependent var	0.000322
Adjusted R-squared	0.115922	S.D. dependent var	0.016680
S.E. of regression	0.015683	Akaike info criterion	-5.461365
Sum squared resid	0.115606	Schwarz criterion	-5.417541
Log likelihood	1302.074	Hannan-Quinn criter.	-5.444131
F-statistic	16.53799	Durbin-Watson stat	1.961088
Prob(F-statistic)	0.000000		

Inverted AR Roots	.75	
Inverted MA Roots	.55-.30i	.55+.30i

(AutoARMA vintage 2)

Table 4

Dependent Variable: DLOG(GDP)
Method: Least Squares
Date: 10/18/25 Time: 16:22
Sample (adjusted): 1980Q3 2019Q2
Included observations: 156 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.002027	0.000439	4.613889	0.0000
DLOG(GDP(-1))	0.204152	0.081927	2.491870	0.0138
DLOG(IPI_R1A)	0.198368	0.023361	8.491592	0.0000
DLOG(IPI_R1A(-1))	0.009061	0.027332	0.331502	0.7407

R-squared	0.503157	Mean dependent var	0.002813
Adjusted R-squared	0.493351	S.D. dependent var	0.006741
S.E. of regression	0.004798	Akaike info criterion	-7.815776
Sum squared resid	0.003500	Schwarz criterion	-7.737575
Log likelihood	613.6305	Hannan-Quinn criter.	-7.784014
F-statistic	51.31051	Durbin-Watson stat	2.061120
Prob(F-statistic)	0.000000		

(BM 1)

Dependent Variable: DLOG(GDP)
Method: Least Squares
Date: 10/18/25 Time: 16:35
Sample (adjusted): 1980Q3 2019Q2
Included observations: 156 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.002026	0.000439	4.611685	0.0000
DLOG(GDP(-1))	0.204596	0.081911	2.497800	0.0136
DLOG(IPI_R2A)	0.198385	0.023356	8.493916	0.0000
DLOG(IPI_R2A(-1))	0.008915	0.027333	0.326184	0.7447

R-squared	0.503219	Mean dependent var	0.002813
Adjusted R-squared	0.493414	S.D. dependent var	0.006741
S.E. of regression	0.004798	Akaike info criterion	-7.815902
Sum squared resid	0.003499	Schwarz criterion	-7.737700
Log likelihood	613.6403	Hannan-Quinn criter.	-7.784140
F-statistic	51.32330	Durbin-Watson stat	2.060947
Prob(F-statistic)	0.000000		

(BM 2)

Dependent Variable: DLOG(GDP)
Method: Least Squares
Date: 10/18/25 Time: 17:18
Sample (adjusted): 1980Q3 2019Q2
Included observations: 156 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.002027	0.000439	4.612250	0.0000
DLOG(GDP(-1))	0.204449	0.081913	2.495941	0.0136
DLOG(IPI_R3)	0.198122	0.023350	8.484705	0.0000
DLOG(IPI_R3(-1))	0.009210	0.027319	0.337123	0.7365

R-squared	0.502895	Mean dependent var	0.002813
Adjusted R-squared	0.493084	S.D. dependent var	0.006741
S.E. of regression	0.004800	Akaike info criterion	-7.815249
Sum squared resid	0.003502	Schwarz criterion	-7.737048
Log likelihood	613.5895	Hannan-Quinn criter.	-7.783487
F-statistic	51.25681	Durbin-Watson stat	2.061070
Prob(F-statistic)	0.000000		

(BM 3)

Table 5: Bridge Model

Table 6: MI_1_ALME

Variable	**Coeff.**	**Std. Err.**	**t-Stat.**	**Prob.**
C	0.001945	0.000474	4.100345	0.0001
DLOG(GDP(-1))	0.271193	0.076319	3.553406	0.0005
SLOPE	0.383082	0.067915	5.640578	0.0000
EXPPDL01	0.887921	0.884986	1.003316	0.3173
EXPPDL02	-0.200638	0.216740	-0.925705	0.3561
R-squared				0.390414
Adj. R-squared				0.374373
S.E. of reg.				0.005324
AIC				-7.601843
DW stat				2.054072
Lag Coeff.				
Lag	**Coeff.**			
0	0.108914			
1	0.144976			
2	0.129192			

Table 7: MI_2_ALME

Variable	**Coeff.**	**Std. Err.**	**t-Stat.**	**Prob.**
C	0.001693	0.000457	3.707420	0.0003
DLOG(GDP(-1))	0.358239	0.068045	5.264748	0.0000
SLOPE	0.372582	0.059930	6.216961	0.0000
EXPPDL01	0.301548	0.767557	0.392867	0.6950
EXPPDL02	-0.128206	0.204897	-0.625708	0.5324
R-squared				0.421614
Adj. R-squared				0.406393
S.E. of reg.				0.005186
AIC				-7.654380
DW stat				2.196118
Lag Coeff.				
Lag	**Coeff.**			
0	0.144655			
1	0.133124			
2	0.094803			

Table 8: MI_3_ALME

Variable	**Coeff.**	**Std. Err.**	**t-Stat.**	**Prob.**
C	0.001678	0.000447	3.755620	0.0002
DLOG(GDP(-1))	0.368950	0.064670	5.705156	0.0000
SLOPE	0.384012	0.055608	6.905752	0.0000
EXPPDL01	1.559161	0.772291	2.018878	0.0453
EXPPDL02	-0.398128	0.181211	-2.197042	0.0295
R-squared				0.441592
Adj. R-squared				0.426897
S.E. of reg.				0.005096
AIC				-7.689531
DW stat				2.275927
Lag Coeff.				
Lag	**Coeff.**			
0	0.113759			
1	0.163835			
2	0.106418			

Table 9: MI_1_PDL1

Variable	**Coeff.**	**Std. Err.**	**t-Stat.**	**Prob.**
C	0.001909	0.000470	4.059556	0.0001
DLOG(GDP(-1))	0.285063	0.074752	3.813433	0.0002
PDL01	0.121657	0.021636	5.622929	0.0000
R-squared				0.385762
Adj. R-squared				0.377785
S.E. of reg.				0.005309
AIC				-7.619718
DW stat				2.064719
Lag Coeff.				
Lag	**Coeff.**			
0	0.121657			
1	0.121657			
2	0.121657			

Table 10: MI_2_PDL1

Variable	**Coeff.**	**Std. Err.**	**t-Stat.**	**Prob.**
C	0.001735	0.000453	3.829235	0.0002
DLOG(GDP(-1))	0.345969	0.066770	5.181486	0.0000
PDL01	0.125776	0.019675	6.392510	0.0000
R-squared				0.414910
Adj. R-squared				0.407311
S.E. of reg.				0.005182
AIC				-7.668333
DW stat				2.189366
Lag Coeff.				
Lag	**Coeff.**			
0	0.125776			
1	0.125776			
2	0.125776			

Table 11: MI_3_PDL1

Variable	**Coeff.**	**Std. Err.**	**t-Stat.**	**Prob.**
C	0.001705	0.000449	3.797743	0.0002
DLOG(GDP(-1))	0.365141	0.064970	5.620121	0.0000
PDL01	0.122016	0.018424	6.622545	0.0000
R-squared				0.423763
Adj. R-squared				0.416280
S.E. of reg.				0.005143
AIC				-7.683581
DW stat				2.269950
Lag Coeff.				
Lag	**Coeff.**			
0	0.122016			
1	0.122016			
2	0.122016			

Table 12: MI_1 PDL2

Variable	Coeff.	Std.Err.	t-Stat.	Prob.
C	0.001918	0.000471	4.068017	0.0001
DLOG(GDP(-1))	0.280992	0.075159	3.738648	0.0003
PDL01	0.101198	0.038378	2.636878	0.0092
PDL02	0.010624	0.016444	0.646029	0.5192
R-squared				0.387433
Adj. R-squared				0.375422
S.E. of reg.				0.005320
AIC				-7.609703
DW stat				2.067790
Lag Coeff.				
Lag	Coeff.			
0	0.111821			
1	0.122445			
2	0.133069			

Table 13: MI_2 PDL2

Variable	Coeff.	Std.Err.	t-Stat.	Prob.
C	0.001690	0.000454	3.721568	0.0003
DLOG(GDP(-1))	0.359699	0.067617	5.319681	0.0000
PDL01	0.170107	0.041419	4.106955	0.0001
PDL02	-0.023495	0.019326	-1.215747	0.2260
R-squared				0.420508
Adj. R-squared				0.409145
S.E. of reg.				0.005174
AIC				-7.665209
DW stat				2.207054
Lag Coeff.				
Lag	Coeff.			
0	0.146612			
1	0.123116			
2	0.099621			

Table 14: MI_3 PDL2

Variable	Coeff.	Std.Err.	t-Stat.	Prob.
C	0.001712	0.000450	3.803458	0.0002
DLOG(GDP(-1))	0.363753	0.065144	5.583835	0.0000
PDL01	0.146105	0.043735	3.340716	0.0011
PDL02	-0.011563	0.019030	-0.607604	0.5444
R-squared				0.425150
Adj. R-squared				0.413879
S.E. of reg.				0.005153
AIC				-7.673252
DW stat				2.286006
Lag Coeff.				
Lag	Coeff.			
0	0.134543			
1	0.122980			
2	0.111418			

Table 15: MI_1 UNRE

Variable	Coeff.	Std.Err.	t-Stat.	Prob.
C	0.001945	0.000473	4.113466	0.0001
DLOG(GDP(-1))	0.271193	0.076076	3.564777	0.0005
Lag1	0.108914	0.026725	4.075301	0.0001
Lag2	0.144976	0.033987	4.265664	0.0000
Lag3	0.129192	0.028345	4.557860	0.0000
R-squared				0.390414
Adj. R-squared				0.374373
S.E. of reg.				0.005324
AIC				-7.601843
DW stat				2.054072

Table 16: MI_2 UNRE

Variable	Coeff.	Std.Err.	t-Stat.	Prob.
C	0.001693	0.000455	3.719236	0.0003
DLOG(GDP(-1))	0.358239	0.067828	5.281565	0.0000
Lag1	0.144655	0.026382	5.483183	0.0000
Lag2	0.133124	0.027150	4.903221	0.0000
Lag3	0.094803	0.030538	3.104437	0.0023
R-squared				0.421614
Adj. R-squared				0.406393
S.E. of reg.				0.005186
AIC				-7.654380
DW stat				2.196118

Table 17: MI_3 UNRE

Variable	Coeff.	Std.Err.	t-Stat.	Prob.
C	0.001678	0.000445	3.767628	0.0002
DLOG(GDP(-1))	0.368950	0.064463	5.723426	0.0000
Lag1	0.113759	0.029076	3.912446	0.0001
Lag2	0.163835	0.026621	6.154282	0.0000
Lag3	0.106418	0.025226	4.218598	0.0000
R-squared				0.441592
Adj. R-squared				0.426897
S.E. of reg.				0.005096
AIC				-7.689531
DW stat				2.275927

Dependent Variable: DLOG(GDP)
Method: MIDAS
Date: 10/18/25 Time: 16:22
Sample (adjusted): 1981Q3 2019Q2
Included observations: 152 after adjustments
Method: Auto/Gets
5 MIDAS terms selected

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.001924	0.000463	4.158201	0.0001
DLOG(GDP(-1))	0.250178	0.075748	3.302768	0.0012
Page: IPI Series: DLOG(IPI 237(-2))				
LAG1	0.110190	0.026689	4.128609	0.0001
LAG2	0.158019	0.035868	4.405521	0.0000
LAG3	0.132322	0.029333	4.510986	0.0000
LAG7	0.054169	0.023726	2.283081	0.0239
LAG16	0.049761	0.023713	2.098493	0.0376

R-squared	0.451423	Mean dependent var	0.002834
Adjusted R-squared	0.428723	S.D. dependent var	0.006794
S.E. of regression	0.005135	Akaike info criterion	-7.660433
Sum squared resid	0.003824	Schwarz criterion	-7.521176
Log likelihood	589.1929	Hannan-Quinn criter.	-7.603862
Durbin-Watson stat	2.064742		

(GETS 1)

Dependent Variable: DLOG(GDP)
Method: MIDAS
Date: 10/18/25 Time: 16:35
Sample (adjusted): 1981Q3 2019Q2
Included observations: 152 after adjustments
Method: Auto/Gets
4 MIDAS terms selected

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.002014	0.000438	4.597530	0.0000
DLOG(GDP(-1))	0.226106	0.071837	3.147486	0.0020
Page: IPI Series: DLOG(IPI 238(-1))				
LAG1	0.135752	0.026421	5.138139	0.0000
LAG2	0.162190	0.026769	6.058952	0.0000
LAG3	0.148434	0.033803	4.391151	0.0000
LAG4	0.126033	0.027382	4.602724	0.0000

R-squared	0.505254	Mean dependent var	0.002834
Adjusted R-squared	0.488311	S.D. dependent var	0.006794
S.E. of regression	0.004860	Akaike info criterion	-7.776875
Sum squared resid	0.003448	Schwarz criterion	-7.657511
Log likelihood	597.0425	Hannan-Quinn criter.	-7.728385
Durbin-Watson stat	2.000969		

(GETS 2)

Dependent Variable: DLOG(GDP)
Method: MIDAS
Date: 10/18/25 Time: 17:18
Sample (adjusted): 1981Q3 2019Q2
Included observations: 152 after adjustments
Method: Auto/Gets
5 MIDAS terms selected

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.002130	0.000418	5.100768	0.0000
DLOG(GDP(-1))	0.160675	0.070227	2.287925	0.0236
Page: IPI Series: DLOG(IPI 239)				
LAG1	0.115313	0.028586	4.033972	0.0001
LAG2	0.157233	0.025739	6.108804	0.0000
LAG3	0.166834	0.025479	6.547810	0.0000
LAG4	0.156701	0.032276	4.854995	0.0000
LAG5	0.111795	0.026332	4.245536	0.0000

R-squared	0.555402	Mean dependent var	0.002834
Adjusted R-squared	0.537005	S.D. dependent var	0.006794
S.E. of regression	0.004623	Akaike info criterion	-7.870592
Sum squared resid	0.003099	Schwarz criterion	-7.731334
Log likelihood	605.1650	Hannan-Quinn criter.	-7.814020
Durbin-Watson stat	1.987642		

(GETS 3)

Table 18: GETS models

Figure 2: Unrestricted Principal Components

Dependent Variable: DLOG(GDP)
Method: MIDAS
Date: 10/18/25 Time: 16:26
Sample (adjusted): 2005Q3 2019Q2
Included observations: 56 after adjustments
Method: U-MIDAS

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000508	0.000537	0.945935	0.3505
DLOG(GDP(-1))	-0.194445	0.153771	-1.264510	0.2142
Page: IPI Series: DLOG(IPI_237(-2)) Lags: 3				
LAG1	0.189844	0.041286	4.598239	0.0001
LAG2	0.121548	0.046459	2.616217	0.0129
LAG3	0.099276	0.039964	2.484120	0.0178
Page: SOFT Series: PC1_SOFT1(-1) Lags: 3				
LAG1	0.009804	0.003501	2.800291	0.0082
LAG2	-0.001315	0.004889	-0.269071	0.7894
LAG3	-0.004899	0.003220	-1.521444	0.1369
Page: SOFT Series: PC2_SOFT1(-1) Lags: 3				
LAG1	-0.002171	0.001098	-1.977300	0.0557
LAG2	0.000825	0.000888	0.929081	0.3590
LAG3	0.000971	0.000715	1.357498	0.1831
Page: SOFT Series: PC3_SOFT1(-1) Lags: 3				
LAG1	-0.000895	0.000748	-1.197043	0.2391
LAG2	-0.000661	0.000576	-1.146119	0.2593
LAG3	0.001129	0.000662	1.704599	0.0969
Page: SOFT Series: PC4_SOFT1(-1) Lags: 3				
LAG1	-0.000175	0.000653	-0.268338	0.7900
LAG2	0.000554	0.000778	0.711474	0.4814
LAG3	-0.000751	0.000653	-1.150848	0.2574
Page: SOFT Series: PC5_SOFT1(-1) Lags: 3				
LAG1	-0.001049	0.000848	-1.236843	0.2242
LAG2	-0.001072	0.000611	-1.754687	0.0878
LAG3	0.000114	0.000677	0.168496	0.8671

R-squared	0.856144	Mean dependent var	-0.000142
Adjusted R-squared	0.780220	S.D. dependent var	0.007187
S.E. of regression	0.003369	Akaike info criterion	-8.275837
Sum squared resid	0.000409	Schwarz criterion	-7.552497
Log likelihood	251.7234	Hannan-Quinn criter.	-7.995400
Durbin-Watson stat	2.203842		

(UNPC 1)

Dependent Variable: DLOG(GDP)
Method: MIDAS
Date: 10/18/25 Time: 17:18
Sample (adjusted): 2005Q2 2019Q2
Included observations: 57 after adjustments
Method: U-MIDAS

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000472	0.000555	0.852021	0.3997
DLOG(GDP(-1))	0.019347	0.154131	0.125521	0.9008
Page: IPI Series: DLOG(IPI_239) Lags: 3				
LAG1	0.085639	0.044870	1.908588	0.0641
LAG2	0.081365	0.047068	1.728682	0.0922
LAG3	0.168354	0.038512	4.371506	0.0001
Page: SOFT Series: PC1_SOFT23 Lags: 3				
LAG1	0.000889	0.004329	0.205401	0.8384
LAG2	0.002722	0.005416	0.502484	0.6183
LAG3	-0.000331	0.003437	-0.096221	0.9239
Page: SOFT Series: PC2_SOFT23 Lags: 3				
LAG1	0.001562	0.001095	1.426078	0.1622
LAG2	-0.000517	0.000976	-0.529205	0.5998
LAG3	-0.000286	0.000705	-0.405576	0.6874
Page: SOFT Series: PC3_SOFT23 Lags: 3				
LAG1	0.000553	0.000752	0.734993	0.4670
LAG2	-0.000705	0.000722	-0.976413	0.3352
LAG3	-0.000688	0.000703	-0.979183	0.3338
Page: SOFT Series: PC4_SOFT23 Lags: 3				
LAG1	0.000442	0.000646	0.684611	0.4979
LAG2	0.000460	0.000641	0.716781	0.4780
LAG3	0.000705	0.000754	0.936196	0.3552
Page: SOFT Series: PC5_SOFT23 Lags: 3				
LAG1	-0.000800	0.000701	-1.141894	0.2608
LAG2	-0.000365	0.000898	-0.406331	0.6868
LAG3	0.000606	0.000750	0.807813	0.4244

R-squared	0.830028	Mean dependent var	3.24E-05
Adjusted R-squared	0.742745	S.D. dependent var	0.007243
S.E. of regression	0.003674	Akaike info criterion	-8.105596
Sum squared resid	0.000499	Schwarz criterion	-7.388736
Log likelihood	251.0095	Hannan-Quinn criter.	-7.827000
Durbin-Watson stat	2.092468		

(UNPC 2)

Dependent Variable: DLOG(GDP)
Method: MIDAS
Date: 10/18/25 Time: 17:18
Sample (adjusted): 2005Q2 2019Q2
Included observations: 57 after adjustments
Method: U-MIDAS

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000472	0.000555	0.852021	0.3997
DLOG(GDP(-1))	0.019347	0.154131	0.125521	0.9008
Page: IPI Series: DLOG(IPI_239) Lags: 3				
LAG1	0.085639	0.044870	1.908588	0.0641
LAG2	0.081365	0.047068	1.728682	0.0922
LAG3	0.168354	0.038512	4.371506	0.0001
Page: SOFT Series: PC1_SOFT23 Lags: 3				
LAG1	0.000889	0.004329	0.205401	0.8384
LAG2	0.002722	0.005416	0.502484	0.6183
LAG3	-0.000331	0.003437	-0.096221	0.9239
Page: SOFT Series: PC2_SOFT23 Lags: 3				
LAG1	0.001562	0.001095	1.426078	0.1622
LAG2	-0.000517	0.000976	-0.529205	0.5998
LAG3	-0.000286	0.000705	-0.405576	0.6874
Page: SOFT Series: PC3_SOFT23 Lags: 3				
LAG1	0.000553	0.000752	0.734993	0.4670
LAG2	-0.000705	0.000722	-0.976413	0.3352
LAG3	-0.000688	0.000703	-0.979183	0.3338
Page: SOFT Series: PC4_SOFT23 Lags: 3				
LAG1	0.000442	0.000646	0.684611	0.4979
LAG2	0.000460	0.000641	0.716781	0.4780
LAG3	0.000705	0.000754	0.936196	0.3552
Page: SOFT Series: PC5_SOFT23 Lags: 3				
LAG1	-0.000800	0.000701	-1.141894	0.2608
LAG2	-0.000365	0.000898	-0.406331	0.6868
LAG3	0.000606	0.000750	0.807813	0.4244

R-squared	0.830028	Mean dependent var	3.24E-05
Adjusted R-squared	0.742745	S.D. dependent var	0.007243
S.E. of regression	0.003674	Akaike info criterion	-8.105596
Sum squared resid	0.000499	Schwarz criterion	-7.388736
Log likelihood	251.0095	Hannan-Quinn criter.	-7.827000
Durbin-Watson stat	2.092468		

(UNPC 3)

Table 19: Principal Components with GETS

Dependent Variable: DLOG(GDP)
Method: MIDAS
Date: 10/29/25 Time: 13:42
Sample (adjusted): 2005Q3 2019Q2
Included observations: 56 after adjustments
Method: Auto/Gets
9 MIDAS terms selected

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000660	0.000474	1.394084	0.1701
DLOG(GDP(-1))	-0.236814	0.139825	-1.693639	0.0972
Page: IPI Series: DLOG(IPI_237(-2))				
LAG1	0.177611	0.036739	4.834393	0.0000
LAG2	0.117137	0.041505	2.822243	0.0071
LAG3	0.099336	0.033072	3.003674	0.0043
Page: SOFT Series: PC1_SOFT1(-1)				
LAG1	0.009741	0.001657	5.878147	0.0000
LAG3	-0.006111	0.001633	-3.742980	0.0005
Page: SOFT Series: PC2_SOFT1(-1)				
LAG1	-0.001776	0.000792	-2.241966	0.0299
LAG3	0.001517	0.000565	2.687507	0.0101
Page: SOFT Series: PC3_SOFT1(-1)				
LAG1	-0.001569	0.000627	-2.503636	0.0160
Page: SOFT Series: PC4_SOFT1(-1)				
Page: SOFT Series: PC5_SOFT1(-1)				
LAG1	-0.001279	0.000668	-1.915104	0.0618

R-squared	0.828482	Mean dependent var	-0.000142
Adjusted R-squared	0.790367	S.D. dependent var	0.007187
S.E. of regression	0.003290	Akaike info criterion	-8.421392
Sum squared resid	0.000487	Schwarz criterion	-8.023556
Log likelihood	246.7990	Hannan-Quinn criter.	-8.267152
Durbin-Watson stat	2.440529		

(GEPC 1)

Dependent Variable: DLOG(GDP)
Method: MIDAS
Date: 10/18/25 Time: 16:35
Sample (adjusted): 2005Q2 2019Q2
Included observations: 57 after adjustments
Method: Auto/Gets
5 MIDAS terms selected

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000378	0.000481	0.785864	0.4357
DLOG(GDP(-1))	-0.178352	0.130440	-1.367309	0.1776
Page: IPI Series: DLOG(IPI 238(-1))				
LAG2	0.219594	0.033037	6.646989	0.0000
LAG3	0.100605	0.038248	2.630343	0.0113
Page: SOFT Series: PC1 SOFT23				
LAG1	0.004732	0.000743	6.369427	0.0000
Page: SOFT Series: PC2 SOFT23				
Page: SOFT Series: PC3 SOFT23				
LAG2	-0.001247	0.000611	-2.042033	0.0464
Page: SOFT Series: PC4 SOFT23				
Page: SOFT Series: PC5 SOFT23				
LAG1	-0.001751	0.000488	-3.590669	0.0008

R-squared	0.786171	Mean dependent var	3.24E-05
Adjusted R-squared	0.760512	S.D. dependent var	0.007243
S.E. of regression	0.003545	Akaike info criterion	-8.332193
Sum squared resid	0.000628	Schwarz criterion	-8.081292
Log likelihood	244.4675	Hannan-Quinn criter.	-8.234684
Durbin-Watson stat	1.594949		

(GEPC 2)

Dependent Variable: DLOG(GDP)
Method: MIDAS
Date: 10/18/25 Time: 17:18
Sample (adjusted): 2005Q2 2019Q2
Included observations: 57 after adjustments
Method: Auto/Gets
8 MIDAS terms selected

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000467	0.000474	0.984820	0.3298
DLOG(GDP(-1))	0.137709	0.109884	1.253221	0.2163
Page: IPI Series: DLOG(IPI 239)				
LAG1	0.109637	0.035108	3.122812	0.0031
LAG2	0.102965	0.035156	2.928837	0.0052
LAG3	0.179938	0.032902	5.468889	0.0000
Page: SOFT Series: PC1 SOFT23				
LAG2	0.002381	0.000830	2.868932	0.0062
Page: SOFT Series: PC2 SOFT23				
LAG1	0.001667	0.000569	2.932219	0.0052
LAG3	-0.001343	0.000568	-2.365309	0.0222
Page: SOFT Series: PC3 SOFT23				
Page: SOFT Series: PC4 SOFT23				
Page: SOFT Series: PC5 SOFT23				
LAG1	-0.000539	0.000575	-0.937464	0.3533
LAG3	0.000865	0.000565	1.531690	0.1323

R-squared	0.804586	Mean dependent var	3.24E-05
Adjusted R-squared	0.767166	S.D. dependent var	0.007243
S.E. of regression	0.003495	Akaike info criterion	-8.316984
Sum squared resid	0.000574	Schwarz criterion	-7.958554
Log likelihood	247.0340	Hannan-Quinn criter.	-8.177686
Durbin-Watson stat	2.161445		

(GEPC 3)

Table 20: Forecasts by model — 2019Q2–2019Q3

	GDP_TARGET	GDP_AUTO	GDP_BM_1	GDP_BM_2	GDP_BM_3
2019Q2	430210.2	403786.1	403786.1	403786.1	403786.1
2019Q3	430459.2	403601.6	404289.3	404227.0	404222.7

GDP_PDL1_1	GDP_PDL2_1	GDP_PDL3_1	GDP_PDL1_2	GDP_PDL2_2	GDP_PDL3_2	GDP_PDL1_3	GDP_PDL2_3	GDP_PDL3_3
403786.1	403786.1	403786.1	403786.1	403786.1	403786.1	403786.1	403786.1	403786.1
404594.1	404666.4	404643.3	404145.1	404190.7	404164.2	404145.8	404163.1	404260.6

	GDP_STEP_1	GDP_STEP_2	GDP_STEP_3	GDP_UNRE_1	GDP_UNRE_2	GDP_UNRE_3
2019Q2	403786.1	403786.1	403786.1	403786.1	403786.1	403786.1
2019Q3	404594.1	404145.1	404145.8	404643.3	404164.2	404260.6

	GDP_ALME_1	GDP_ALME_2	GDP_ALME_3	GDP_GETS_1	GDP_GETS_2	GDP_GETS_3
2019Q2	403786.1	403786.1	403786.1	403786.1	403786.1	403786.1
2019Q3	404643.3	404164.2	404260.6	404796.5	404600.2	404484.9

	GDP_UNPC_1	GDP_UNPC_2	GDP_UNPC_3	GDP_GEPC_1	GDP_GEPC_2	GDP_GEPC_3
2019Q2	403786.1	403786.1	403786.1	403786.1	403786.1	403786.1
2019Q3	404433.1	403624.4	403963.3	404398.0	403194.0	403488.2

Table 21: Percentage Growth (2019Q2–2019Q3)

	GDP_TARGET	GDP_AUTO	GDP_BM_1	GDP_BM_2	GDP_BM_3
2019Q2	0.06%	0.03%	0.03%	0.03%	0.03%
2019Q3	0.06%	-0.05%	0.12%	0.11%	0.11%

GDP_PDL1_1	GDP_PDL2_1	GDP_PDL3_1	GDP_PDL1_2	GDP_PDL2_2	GDP_PDL3_2	GDP_PDL1_3	GDP_PDL2_3	GDP_PDL3_3
0.03%	0.03%	0.03%	0.03%	0.03%	0.03%	0.03%	0.03%	0.03%
0.20%	0.22%	0.21%	0.09%	0.10%	0.12%	0.09%	0.09%	0.09%

	GDP_STEP_1	GDP_STEP_2	GDP_STEP_3	GDP_UNRE_1	GDP_UNRE_2	GDP_UNRE_3
2019Q2	0.03%	0.03%	0.03%	0.03%	0.03%	0.03%
2019Q3	0.20%	0.09%	0.09%	0.21%	0.09%	0.12%

	GDP_ALME_1	GDP_ALME_2	GDP_ALME_3	GDP_GETS_1	GDP_GETS_2	GDP_GETS_3
2019Q2	0.03%	0.03%	0.03%	0.03%	0.03%	0.03%
2019Q3	0.21%	0.09%	0.12%	0.25%	0.20%	0.17%

	GDP_UNPC_1	GDP_UNPC_2	GDP_UNPC_3	GDP_GEPC_1	GDP_GEPC_2	GDP_GEPC_3
2019Q2	0.03%	0.03%	0.03%	0.03%	0.03%	0.03%
2019Q3	0.16%	-0.04%	0.04%	0.15%	-0.15%	-0.07%